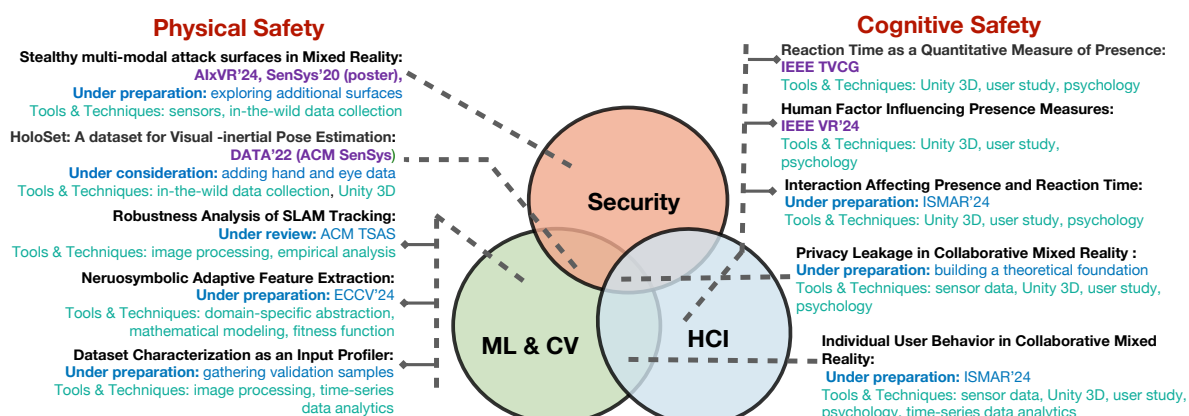


Towards Secure and Immersive Mixed Reality

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In recent years, the evolution of Mixed Reality (MR) has taken a remarkable turn, transitioning from primarily recreational use to becoming an indispensable tool in critical applications such as surgery and therapy. While these technological advancements have undoubtedly enhanced human lives, there is a growing concern that the rapid pace of change might outstrip our understanding of the potential consequences a concern rooted in the lessons learned from past technological adoptions. In my research, I uncover the fundamental challenges in MR platforms and devices, focusing on the potential sources of physical and cognitive discomfort for users. This exploration unfolds along two critical dimensions, each revealing a complex interplay between technology and human experience.

Physical Safety: Delving into physical safety, I explore attack surfaces on MR platforms that expose users to a wide range of attacks that harm their physical safety despite state-of-the-art mitigation techniques and system checks. My work focuses on a new attack surface resilient to sensor-fusion techniques as it concurrently manipulates multiple sensing streams across spatiotemporal axes (published at **AIxVR'24**). This work set the stage for a deeper examination of flaws in Simultaneous Localization and Mapping (SLAM) tracking, a core component of MR systems. My ongoing research proposes strategies for enhancing the robustness of SLAM. A forthcoming paper outlines a novel approach to addressing internal (errors) and external (malicious) tracking vulnerabilities. This work uses a neurosymbolic domain-specific language to create abstractions for adaptive feature extraction, a tailored solution to fortify MR systems against potential physical security threats and system errors. Additionally, to bolster the understanding of MR systems, a novel dataset named HoloSet was introduced (published at **DATA'22**). HoloSet, captured using Microsoft Hololens 2, provides $\approx 100K$ samples of data with visual, inertial, and depth camera frames for macro and micro-movements in various environments.



Cognitive Safety: The quest for cognitive safety within MR environments has been equally rigorous, exploring the nuanced dynamics of user interaction and experience. Our journey started with a fundamental question: Does an individual experiencing heightened *presence* in MR environments consistently exhibit faster reaction times? This inquiry seeks to provide a systemic metric, reaction time, that objectively quantifies presence in a non-intrusive and unbiased manner. The exploration of this question began with an investigation into technical factors, including the appearance and behavior of objects, culminating in an accepted paper in **IEEE TVCG**. Subsequently, we delved into human factors, focusing on conditioning (paper accepted at **IEEE VR'24**). The ongoing trajectory of this research involves understanding how user interactions fit into the broader picture of presence and its impact on cognitive safety. We are further working on biometric profiling of users in collaborative MR. We are identifying metrics to create the biometric profile of users, leveraging experience measures, performance, perceptual responses, and sensory data to identify the user behavior and possible privacy leakages.

As my research narrative unfolds, the threads of physical and cognitive safety remain intertwined, painting a holistic picture of the challenges and opportunities presented by MR technology. My research seeks to identify design challenges and pave the way for innovative solutions, ensuring that the integration of MR into our lives enhances experiences without compromising safety and well-being of humans that are at the center of this technology.